program 2

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2.Write an R script to create a scatter plot, incorporating categorical analysis through color-coded data points representing different groups, using ggplot2.

Step 1: Load necessary libraries

```
library(ggplot2)
```

Step 2: Load the Dataset

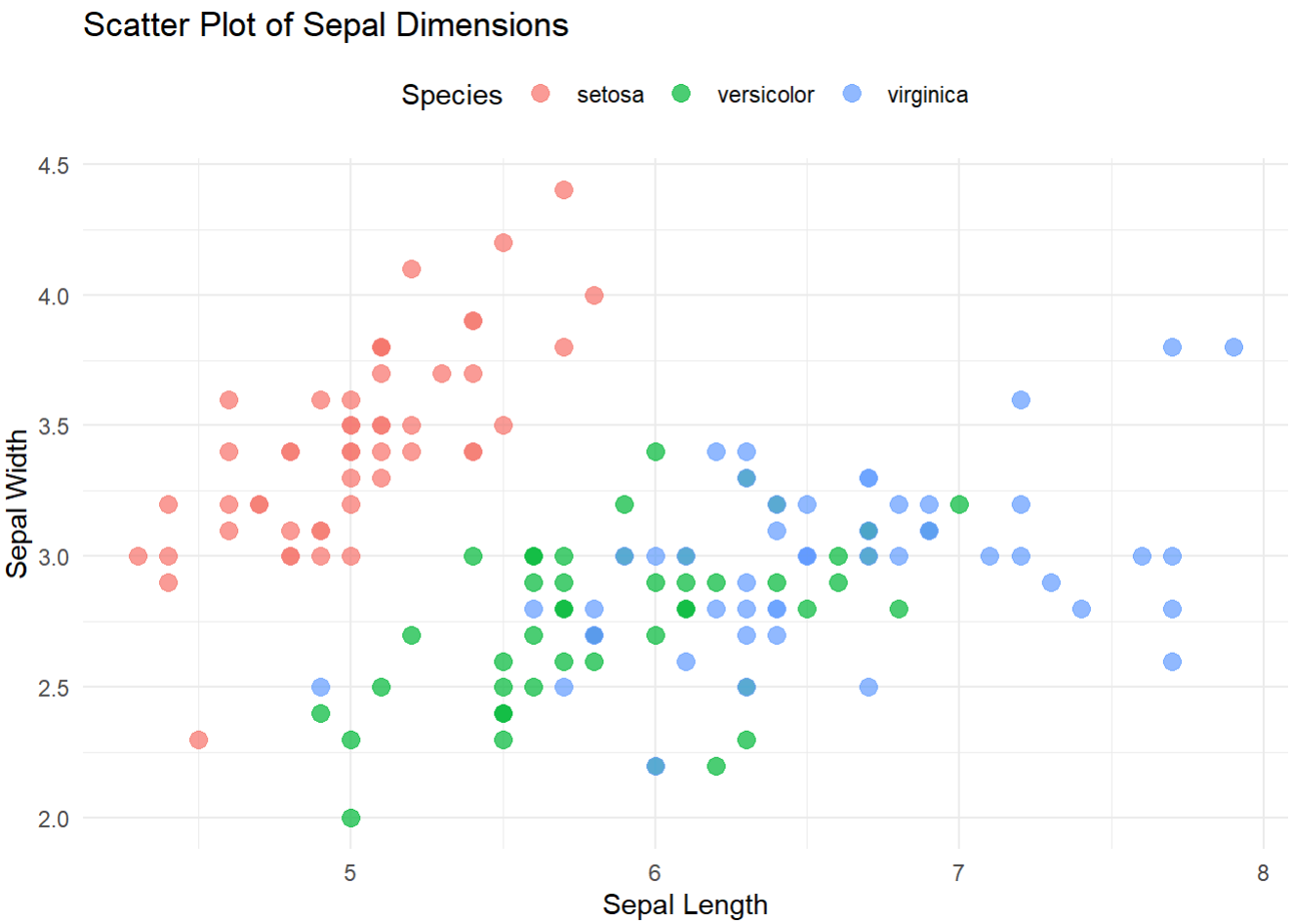
```
# Load the iris dataset
data <- iris

# Display first few rows
head(data)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

Step 3: Create a Scatter Plot

```
# Create a scatter plot using ggplot2
ggplot(data, aes(x = Sepal.Length, y = Sepal.Width, color = Species)) +
  geom_point(size = 3, alpha = 0.7) +  # Increase point size & transparency
  labs(title = "Scatter Plot of Sepal Dimensions",
       x = "Sepal Length",
       y = "Sepal Width",
       color = "Species") +  # Legend title
  theme_minimal() +  # Clean layout
  theme(legend.position = "top")  # Move legend to the top
```


Scatter Plot of Sepal Dimensions
Species
setosa
versicolor
virginica
Sepal Width
4.5
4.0
3.5
3.0
2.5
2.0
5
6
7
8
Sepal Length